Homotopy

from topology to homological algebra

The notion of homotopy appears in many places, most naturally when studying topological spaces, but also in more general and algebraic context, as we will briefly review in this note, closely following [1].

1 Homotopies between topological spaces

Let us start with the definition of an homotopy between continuous maps relating two topological spaces.

Definition 1.1 [Homotopy]**.** Let X and Y be two topological spaces, and $f, g : X \to Y$ two continuous maps between them. A homotopy between f and g, sometimes denoted by

$$X \;\underset{g}{\overset{f}{\Downarrow H}}\; Y \tag{1.1}$$

is a continuous map

$$H : [0,1] \times X \longrightarrow Y\,, \tag{1.2}$$

such that $H(0,-) = f$ and $H(1,-) = g$.

Homotopies can be composed in two different ways:

(i) Consider the situation illustrated below,

$$X \;\underset{g_1}{\overset{f_1}{\Downarrow H_1}}\; Y \;\underset{g_2}{\overset{f_2}{\Downarrow H_2}}\; Z \tag{1.3}$$

where we have two pairs of continuous maps

$$f_1, g_1 : X \longrightarrow Y \qquad \text{and} \qquad f_2, g_2 : Y \longrightarrow Z\,,$$

which are related by two homotopies

$$H_1 : [0,1] \times X \longrightarrow Y \qquad \text{and} \qquad H_2 : [0,1] \times Y \longrightarrow Z\,.$$

A natural question to ask, in this situation, is whether it is possible to define an homotopy between the composition of each pairs, i.e.

$$X \;\underset{g_2 \circ g_1}{\overset{f_2 \circ f_1}{\Downarrow H_2 \circ_h H_1}}\; Z \qquad \text{with} \qquad H_2 \circ_h H_1 : [0,1] \times X \longrightarrow Z\,, \tag{1.4}$$

constructed out of the two existing homotopies H_1 and H_2? The answer is positive, and given by

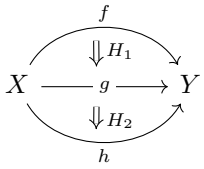
$$(H_2 \circ_h H_1)(t,x) := H_2\big(t, H_1(t,x)\big)\,, \tag{1.5}$$

called the *horizontal composition* of H_1 and H_2. As the composition of continuous maps, it is also continuous, and clearly verifies

$$(H_2 \circ_h H_1)(0,-) = f_2 \circ f_1(-) \qquad \text{and} \qquad (H_2 \circ_h H_1)(1,-) = g_2 \circ g_1(-)\,, \tag{1.6}$$

and hence does define an homotopy between the compositions $f_2 \circ f_1$ and $g_2 \circ g_1$.

(*ii*) Now let us turn to another type of situation,



(1.7)

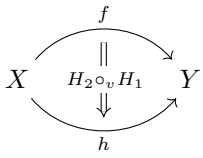
where we three contiunuous maps

$$f, g, h : X \longrightarrow Y \,,$$

between the same spaces, and related pair by pair by two homotopies,

$$H_1, H_2 : [0,1] \times X \longrightarrow Y \,.$$

As in the previous case, there is an obvious question that poses itself here, which whether we can define an homotopy between f and h,



with $\qquad H_2 \circ_v H_1 : [0,1] \times X \longrightarrow Y \,,$ (1.8)

from the original two homotopies H_1 and H_2. The answer, positive again, is given by

$$(H_2 \circ_v H_1)(t,x) := \begin{cases} H_1(2t,x) & 0 \leq t \leq \frac{1}{2} \,, \\ H_2(2t-1,x) & \frac{1}{2} \leq t \leq 1 \,, \end{cases} \tag{1.9}$$

which is called the *vertical composition* of H_1 and H_2.

With this last notion of composition at hand, we are now in position to prove the following proposition.

Proposition 1.1. *The property that two maps are homotopic to one another is an equivalence relation.*

Proof. It is clear that any continuous map is homotopic to itself, i.e. being homotopic is a reflexive relation. Moreover, if $H : f \Longrightarrow g$ is an homotopy from f to g, then

$$H^{-1}(t,-) := H(1-t,-) \,, \tag{1.10}$$

defines an homotopy from g to f, so that being homotopic is a symmetric relation. Finally, since for any three continuous maps $f, g, h : X \longrightarrow Y$ which are related by $H_1 : f \Longrightarrow g$ and $H_2 : g \Longrightarrow h$, the vertical composition $H_2 \circ_v H_1 : f \Longrightarrow h$ defines an homotopy between f and h, being homotopic is also a transitive relation. ■

(∗) Remark 1.1. It is clear that the horizontal composition of homotopies is *associative*, i.e.

$$(H_3 \circ_h H_2) \circ_h H_1 = H_3 \circ_h (H_2 \circ_h H_1) \,, \tag{1.11}$$

for any triple of homotopies H_1, H_2 and H_3 (which are of course composable horizontally). The vertical composition of homotopies is, however, *not associative*. Explicitly, the two possible compositions above read

$$(H_3 \circ_v H_2) \circ_v H_1 \qquad\qquad H_3 \circ_v (H_2 \circ_v H_1) \tag{1.12}$$

$$= \begin{cases} H_1(2t,x) & 0 \leq t \leq \frac{1}{2} \,, \\ H_2(4t-2,x) & \frac{1}{2} \leq t \leq \frac{3}{4} \,, \\ H_3(4t-3) & \frac{3}{4} \leq t \leq 1 \,, \end{cases} \qquad = \begin{cases} H_1(4t,x) & 0 \leq t \leq \frac{1}{4} \,, \\ H_2(4t-1,x) & \frac{1}{4} \leq t \leq \frac{1}{2} \,, \\ H_3(2t-1) & \frac{1}{2} \leq t \leq 1 \,, \end{cases} \tag{1.13}$$

and hence define two different maps $[0,1] \times X \longrightarrow Y$, i.e. two different homotopies between the composition of two triplets of maps starting at X and ending in Y. The discrepancy between them

is however fairly mild: they only differ by the speed at which the interval $[0,1]$, split in three pieces in two different manner, is traveled. It is therefore not surprising that these two compositions can be related by introducing a parameter controlling the speed at which each piece of the unit interval is traveled, say s, such that for $s=0$ it corresponds to the first case, $(H_3 \circ_v H_2) \circ_v H_1$, and for $s=1$, to the second case, $H_3 \circ_v (H_2 \circ_v H_1)$. This is realised concretely by defining

$$G(s;t,x) = \begin{cases} H_1\big(2(s+1)t, x\big) & 0 \leq t \leq \frac{2-s}{4}\,, \\ H_2\big(2(2t-1)+s, x\big) & \frac{2-s}{4} \leq t \leq \frac{3-s}{4}\,, \\ H_3\big(2(2-s)(t-1)+1, x\big) & \frac{3-s}{4} \leq t \leq 1\,, \end{cases} \tag{1.14}$$

which verifies

$$G(0;-,-) = (H_3 \circ_v H_2) \circ_v H_1\,, \qquad \text{and} \qquad G(1;-,-) = H_3 \circ_v (H_2 \circ_v H_1)\,. \tag{1.15}$$

In other words,

$$G : [0,1] \times \big([0,1] \times X\big) \longrightarrow Y\,, \tag{1.16}$$

defines a *homotopy* between the two homotopies $(H_3 \circ_v H_2) \circ_v H_1$ and $H_3 \circ_v (H_2 \circ_v H_1)$, and hence the vertical composition is associative *only up to homotopy*. This is however enough for homotopies to define a relation of equivalence on continuous maps between topological spaces, as we saw in Proposition 1.1.

2 Algebraic setting

The assignment which sends a manifold M to its complex of forms Ω_M, and a smooth map between manifolds $f : M \to N$ to its pullback $f^* : \Omega_M \to \Omega_N$, defines a *contravariant* functor

$$\mathsf{Mfd} \xrightarrow{\ \Omega\ } \mathsf{dgComAlg}\,, \tag{2.1}$$

between the category of manifolds and that of differential graded commutative algebras. We can therefore transfer the notion of homotopy from topological spaces to differential forms. In order to reformulate the boundary conditions, let us introduce the inclusion

$$\mathrm{ev}_{t_0} : \{t_0\} \hookrightarrow [0,1]\,, \tag{2.2}$$

of any real number $0 \leq t_0 \leq 1$, as the corresponding point on the unit interval. The 'initial and final conditions' for an homotopy $H : [0,1] \times X \longrightarrow Y$ between the pair $f, g : X \longrightarrow Y$ can be written as

$$(\mathrm{ev}_0 \times \mathbf{1}_X)^* H = f\,, \qquad \text{and} \qquad (\mathrm{ev}_1 \times \mathbf{1}_X)^* H = g\,, \tag{2.3}$$

thanks to this evaluation map.

Let $\mathbb{R}[t, \mathrm{d}t]$ denote the differential graded commutative algebra of polynomial in one degree 0 variable t and one degree 1 variable $\mathrm{d}t$, i.e. the algebra of polynomial forms on $\mathbb{R}$.

Definition 2.1 [**Homotopy between chain maps**]**.** Let (C, δ_C) and (D, δ_D) be two complexes, and $f, g : C \to D$ two chain maps between them. A homotopy between f and g is a chain map

$$H : C \longrightarrow \mathbb{R}[t, \mathrm{d}t] \otimes D\,, \tag{2.4}$$

such that

$$(\mathrm{ev}_0 \otimes \mathbf{1})\, H = f\,, \qquad (\mathrm{ev}_1 \otimes \mathbf{1})\, H = g\,. \tag{2.5}$$

Whenever such a homotopy exists, we will say that f and g are homotopic to one another, and will write $f \underset{H}{\sim} g$.

As the notation suggests, the existence of an homotopy between two chain maps is also an equivalence relation.

Lemma 2.1. *Two chain maps $f, g : C \to D$ are homotopic if and only if there exists a degree -1 linear map $h : C \to D$ such that $\delta_D\, h + h\, \delta_C = g - f$.*

Proof. Let $H : C \to \mathbb{R}[t, \mathrm{d}t] \otimes D$ be a homotopy between f and g, then

$$h := \Big(\int_0^1 (-) \otimes \mathbf{1}_D \Big) \circ H \,, \tag{2.6}$$

defines a homotopy operator since we have

$$\delta_D \, h + h \, \delta_C = \Big(\int_0^1 (-) \otimes \mathbf{1}_D \Big) \circ \Big[- \big(\mathbf{1}_{\mathbb{R}[t,\mathrm{d}t]} \otimes \delta_D \big) \circ H + \circ H \circ \delta_C \Big] \,, \tag{2.7}$$

where the minus sign before the first term comes from the fact that both $\int_0^1 (-)$ and δ_D are of odd degrees, so that when the latter generates a sign when 'jumping above' the integral. Now the fact that H is a chain map means that it verifies

$$H \circ \delta_C = \big(\delta_{\mathbb{R}[t,\mathrm{d}t]} \otimes \mathbf{1}_D + \mathbf{1}_{\mathbb{R}[t,\mathrm{d}t]} \otimes \delta_D \big) \circ H \,, \tag{2.8}$$

we can further simplifies the original expression as

$$\delta_D \, h + h \, \delta_C = \Big(\int_0^1 \delta_{\mathbb{R}[t,\mathrm{d}t]} (-) \otimes \mathbf{1}_D \Big) \circ H = g - f \,, \tag{2.9}$$

where we used Stokes' theorem in the last step. Conversely, given an homotopy operator $h : C \to D$, the formula

$$H := t \, g + (1 - t) \, f + \mathrm{d}t \, h \tag{2.10}$$

defines an homotopy between f and g. Indeed, one can check by direct computation that H as defined above is a chain map,

$$\big(\delta_D \otimes \mathbf{1}_{\mathbb{R}[t,\mathrm{d}t]} + \mathbf{1}_D \otimes \delta_{\mathbb{R}[t,\mathrm{d}t]} \big) \circ H = H \circ \delta_C + \mathrm{d}t \, \big(g - f - \delta_D \, h - h \, \delta_C \big) = H \circ \delta_C \,, \tag{2.11}$$

as a consequence of the fact that h is a homotopy operator between f and g. Finally, it is clear that the homotopy operator defined from this homotopy as explained above is nothing but h itself, thereby proving the equivalence between the two notions. ■

Corollary 2.2. *Two homotopic chain maps induce the same map in cohomology.*

Proof. This directly follows from the fact that the difference between two homotopic chain maps vanishes on cohomology: take a cocycle $c \in \mathrm{Ker}(\delta_C) \subset C$, representative of the cohomology class $[c] \in H(\delta_C)$, then

$$g(c) - f(c) = \delta_D \, h(c) \in \mathrm{Im}(\delta_D) \,, \tag{2.12}$$

since $\delta_C c = 0$, which implies that $g(c)$ and $f(c)$ belong to the same cohomology class of δ_D, i.e. $[g(c)] = [f(c)] \in H(\delta_D)$. ■

If the identity map on a complex (C, δ) is homotopic to the zero map, the homotopy operator $h : C \longrightarrow C$ relating them,

$$\delta \, h + h \, \delta = \mathbf{1}_C \,, \tag{2.13}$$

is called a *contracting homotopy*. In light of the previous corollary, the existence of a contracting homotopy implies that the complex has no cohomology, i.e. (C, δ) is acyclic. Indeed, the above relation can be read as the fact that the identity map on C is homotopic to the trivial map, so that by Corollary 2.2, the cohomology of C is the same that of trivial complex, and hence is empty. We record this result in the following lemma.

Lemma 2.3. *Let (C, δ) be a cochain complex, and assume that the identity $\mathbf{1}_C$ is homotopic to a chain map $\sigma : C \longrightarrow C$. Then the kernel of σ is acyclic, and hence $H(\delta) \subseteq C/\mathrm{Ker}(\sigma)$. In particular, if there exists a contracting homotopy, then C is acyclic, $H(\delta) = 0$.*

Proof. This follows from a simple calculation: Take a cocycle $\alpha \in C$, $\delta\alpha = 0$, and apply the homotopy operator h relating $\mathbf{1}_C$ and σ,

$$0 = h \, \delta\alpha = -\delta \, h\alpha + \alpha - \sigma(\alpha) \,, \tag{2.14}$$

so that

$$\sigma(\alpha) = 0 \qquad \Longrightarrow \qquad \alpha = \delta \, h\alpha \,, \tag{2.15}$$

i.e. α is also a coboundary if it belong to the kernel of σ. ■

References

[1] E. Meinrenken, *Clifford algebras and Lie theory*, vol. 58. Springer Berlin, Heidelberg, 2013.